Logs traces and privacy

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete privacy-aware diagnostic record.

Keep the permitted schema, omitted content, trace purpose, span scope and control requirements together.

Keep one habit: Recall → W1–W3/W5 → independent field selection → project diagnostic policy → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Integration tests and fixtures. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Least privilege and tool scopes. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- Logs support diagnosis, but excessive recording can expose private content. **Trace ID**s link stages without storing complete sensitive **input**s.

- For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers. Restrict access and declare retention.

Failure budgets and service objectives

> **My remembered idea, example and correction**
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Integration tests and fixtures

> **My remembered idea, example and correction**
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Least privilege and tool scopes

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main logging policy allows trace ID, stage, status, duration and non-sensitive error category; excludes passcodes, raw student text and teacher answers. Main R7 denied event should keep controlled metadata, not complete passcode. Local mock supplies stage authorize, duration 3 ms, error permission_denied; no actual timing observed. Build a new explicit record and never print/serialize the raw placeholder mock.

  Retain a policy-compliant diagnostic artifact rather than a private raw record. Separate picture trace R17 links request/retrieval/validation; its explicit retrieval policy permits source C8. Request 240 ms, lookup 80 ms, validation 20 ms are supplied mock spans, which can overlap/nest; 240is one request value, not a service average. No timer or live log operation occurs.

  Field/value checks do not establish anonymity: trace/source IDs can be linkable. Access by designated classroom reviewer and deletion after exercise are declared requirements, not implemented controls. No real credentials or private records used.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main logging policy allows trace ID, stage, status, duration and non-sensitive error category; excludes passcodes, raw student text and teacher answers. Main R7 denied event should keep controlled metadata, not complete passcode. Local mock supplies stage authorize, duration 3 ms, error permission_denied; no actual timing observed. Build a new explicit record and never print/serialize the raw placeholder mock.

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  Field/value checks do not establish anonymity: trace/source IDs can be linkable. Access by designated classroom reviewer and deletion after exercise are declared requirements, not implemented controls. No real credentials or private records used.

- A failure record includes trace R7, status=denied and the complete student passcode. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main logging policy allows trace ID, stage, status, duration and non-sensitive error category; excludes passcodes, raw student text and teacher answers. Main R7 denied event should keep controlled metadata, not complete passcode. Local mock supplies stage authorize, duration 3 ms, error permission_denied; no actual timing observed. Build a new explicit record and never print/serialize the raw placeholder mock.

  Retain a policy-compliant diagnostic artifact rather than a private raw record. Separate picture trace R17 links request/retrieval/validation; its explicit retrieval policy permits source C8. Request 240 ms, lookup 80 ms, validation 20 ms are supplied mock spans, which can overlap/nest; 240is one request value, not a service average. No timer or live log operation occurs.

  Field/value checks do not establish anonymity: trace/source IDs can be linkable. Access by designated classroom reviewer and deletion after exercise are declared requirements, not implemented controls. No real credentials or private records used.

- A report omits this qualification: A **trace ID** can still be linkable; **minimisation** and role-controlled access remain necessary. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main logging policy allows trace ID, stage, status, duration and non-sensitive error category; excludes passcodes, raw student text and teacher answers. Main R7 denied event should keep controlled metadata, not complete passcode. Local mock supplies stage authorize, duration 3 ms, error permission_denied; no actual timing observed. Build a new explicit record and never print/serialize the raw placeholder mock.

  Retain a policy-compliant diagnostic artifact rather than a private raw record. Separate picture trace R17 links request/retrieval/validation; its explicit retrieval policy permits source C8. Request 240 ms, lookup 80 ms, validation 20 ms are supplied mock spans, which can overlap/nest; 240is one request value, not a service average. No timer or live log operation occurs.

  Field/value checks do not establish anonymity: trace/source IDs can be linkable. Access by designated classroom reviewer and deletion after exercise are declared requirements, not implemented controls. No real credentials or private records used.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Logs contain user essay, teacher guide, stage=lookup, duration=12 ms and trace T8. Select allowed fields.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled logs traces and privacy **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of logs traces and privacy.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Logs traces and privacy. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Use the worked case for Logs traces and privacy. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Logs contain user essay, teacher guide, stage=lookup, duration=12 ms and trace T8. Select allowed fields.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

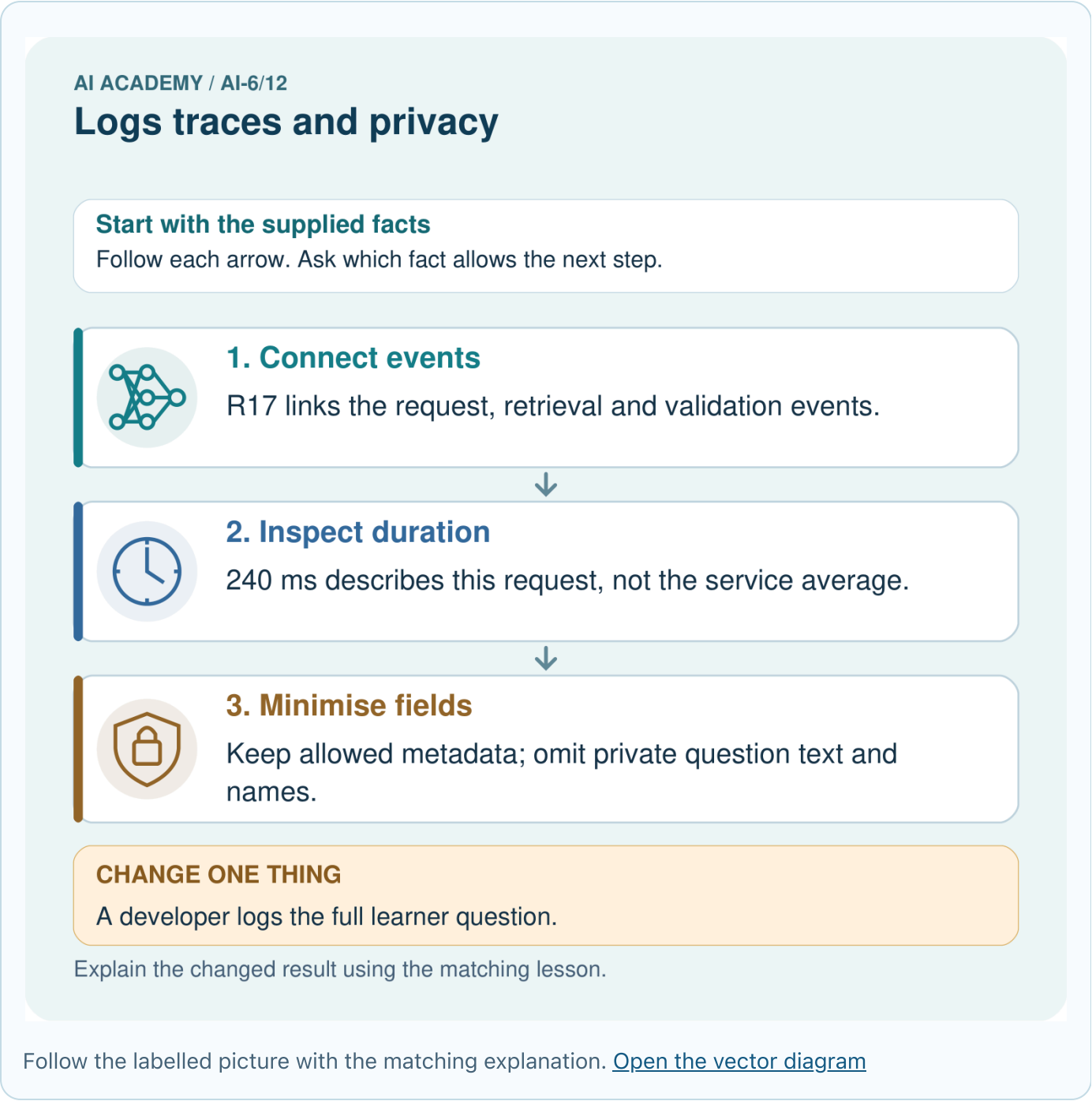
- Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.

- R17 is the request **trace ID**; C8 is the retrieved source ID. Allowed fields also include validation pass and duration 240 ms. Raw question text is excluded.

- Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17.
- Which labelled step matches this detail? Keep allowed metadata; omit private question text and names.
- 1. Inspect duration
- 2. Connect events

- 3. Minimise fields

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check after the first attempt

After your attempt, compare with the supported explanation: Minimise fields Keep allowed metadata; omit private question text and names. Logs record events; traces connect related events.

Useful debugging fields should remain within the declared privacy policy.

Changed-case question and optional hint

- Changed-case practice: A developer logs the full learner question. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. That additional field violates this logging policy.

- 3. This one example guarantees correct results for every future **input**.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: That additional field violates this logging policy. Debugging convenience does not authorise storing more personal information.

Repair and retain the record

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

Review Integration tests and fixtures (ai-6/9). Goal: Explain the **mechanism** and terminology of integration tests and fixtures. Short practice: Follow request N through the fixture and adapter.

Which source and item should appear in the reply, and what precisely fails when the adapter returns M?. Return to this week and explain what you changed.

Review Least privilege and tool scopes (ai-6/6). Goal: Explain the **mechanism** and terminology of least privilege and tool scopes. Short practice: Classify read S2, write S2 and read S5 as permitted or denied.

Identify the operation or station mismatch for each denial, then reconsider after read S5 is added to trusted scope.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

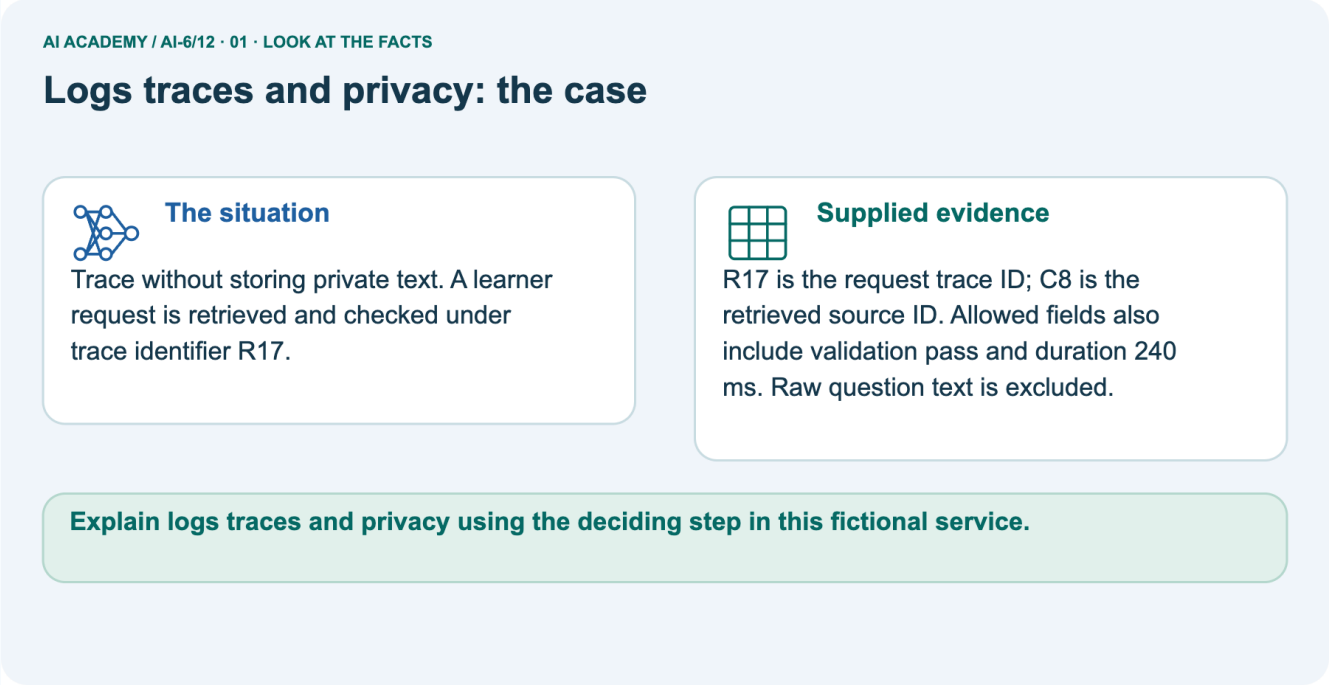
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Logs traces and privacy: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

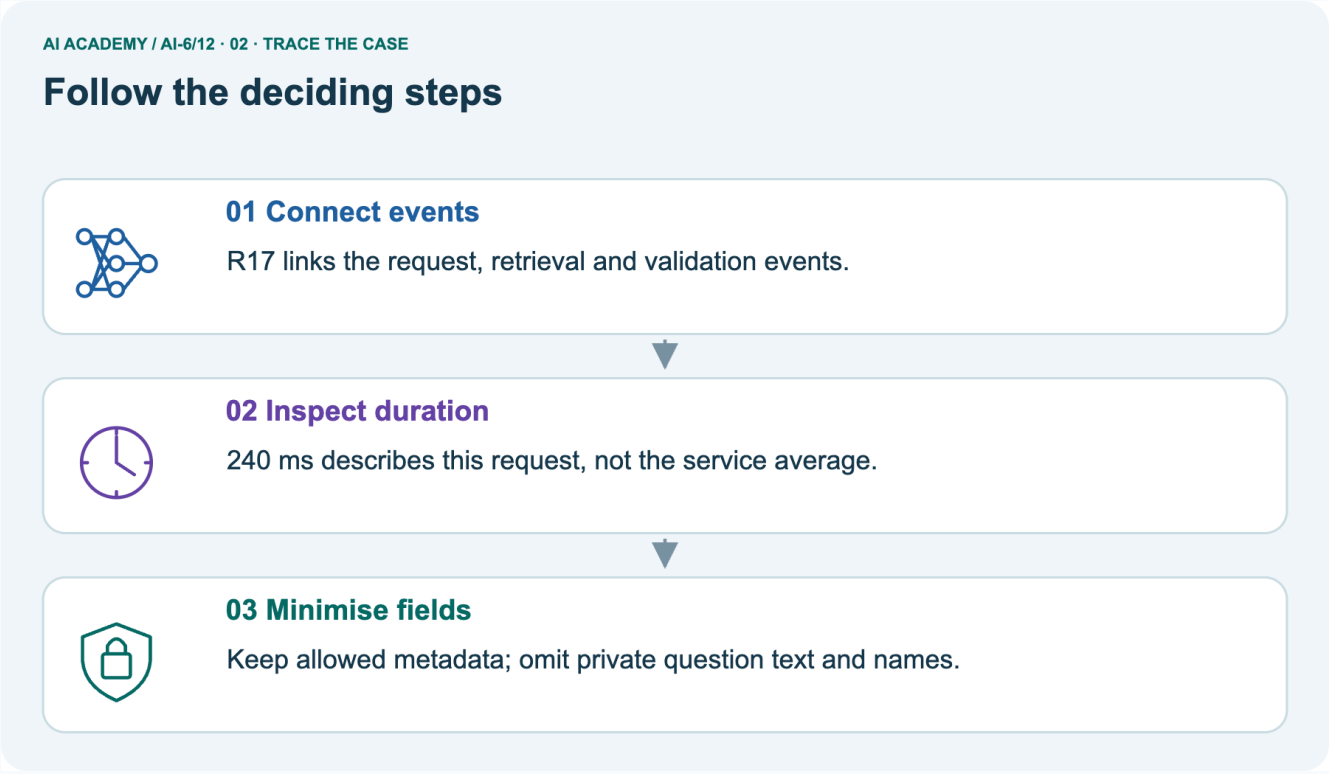
Trace without storing private text. A learner request is retrieved and checked under **trace id**entifier R17. Supplied facts: R17 is the request **trace ID**; C8 is the retrieved source ID.

Allowed fields also include validation pass and duration 240 ms. Raw question text is excluded.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



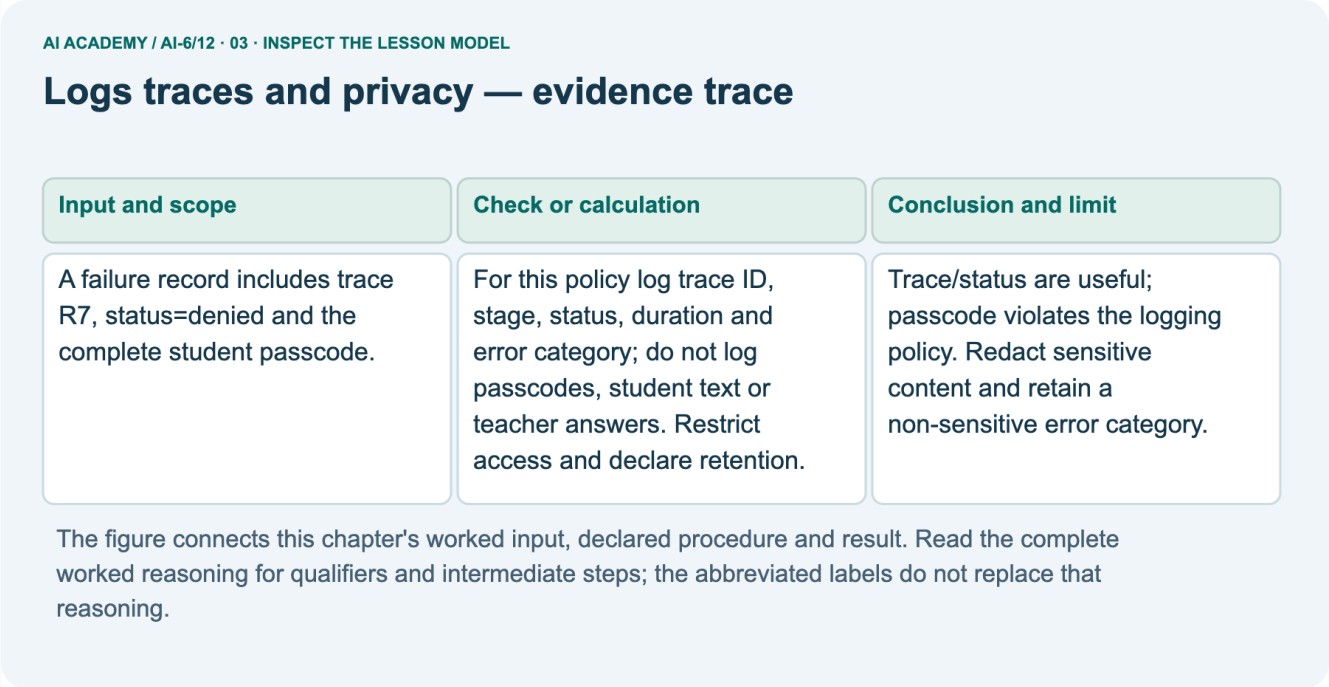
Follow the labelled picture with the matching explanation. Open the vector diagram

Connect events: R17 links the request, retrieval and validation events. → Inspect duration: 240 ms describes this request, not the service average. → Minimise fields: Keep allowed metadata; omit private question text and names.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Logs traces and privacy — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Logs traces and privacy — **evidence** trace.

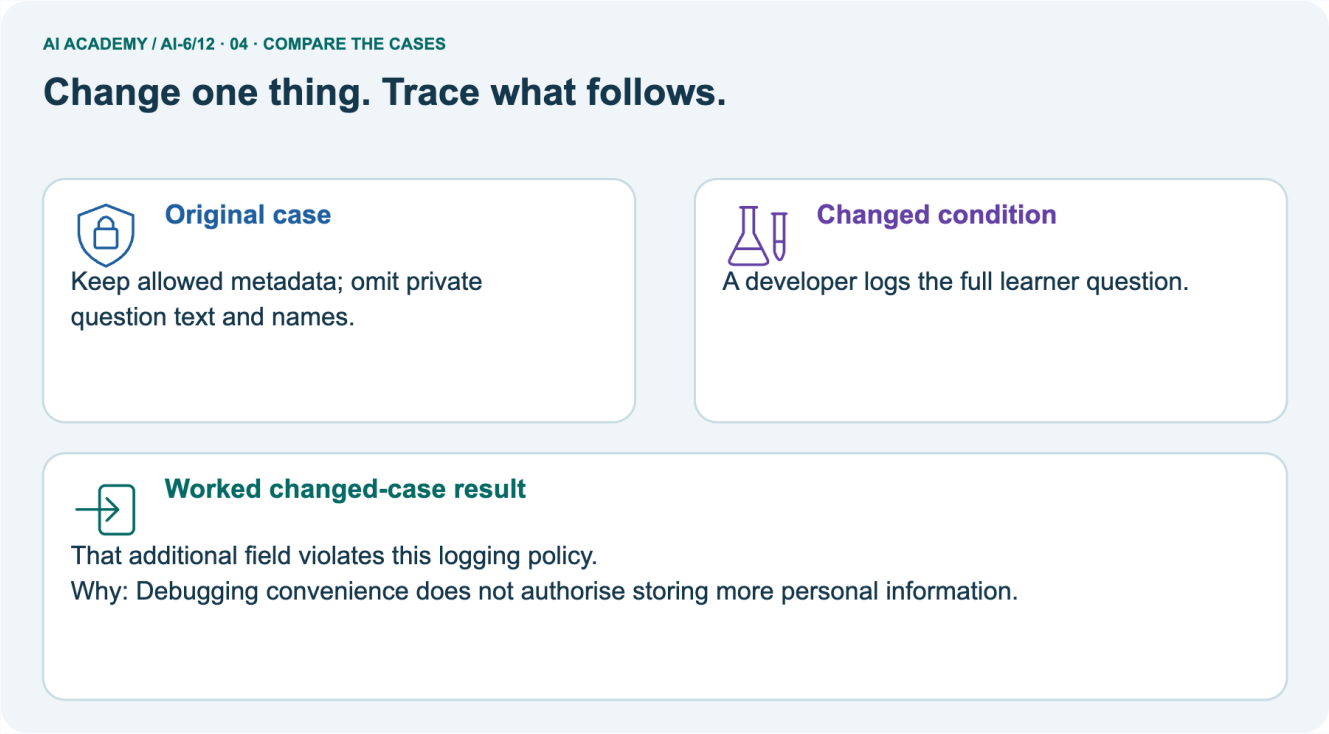
A failure record includes trace R7, status=denied and the complete student passcode. | For this policy log **trace ID**, stage, status, duration and **error category**; do not log passcodes, student text or teacher answers.

Restrict access and declare retention. | Trace/status are useful; passcode violates the logging policy. Redact sensitive content and retain a non-sensitive **error category**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

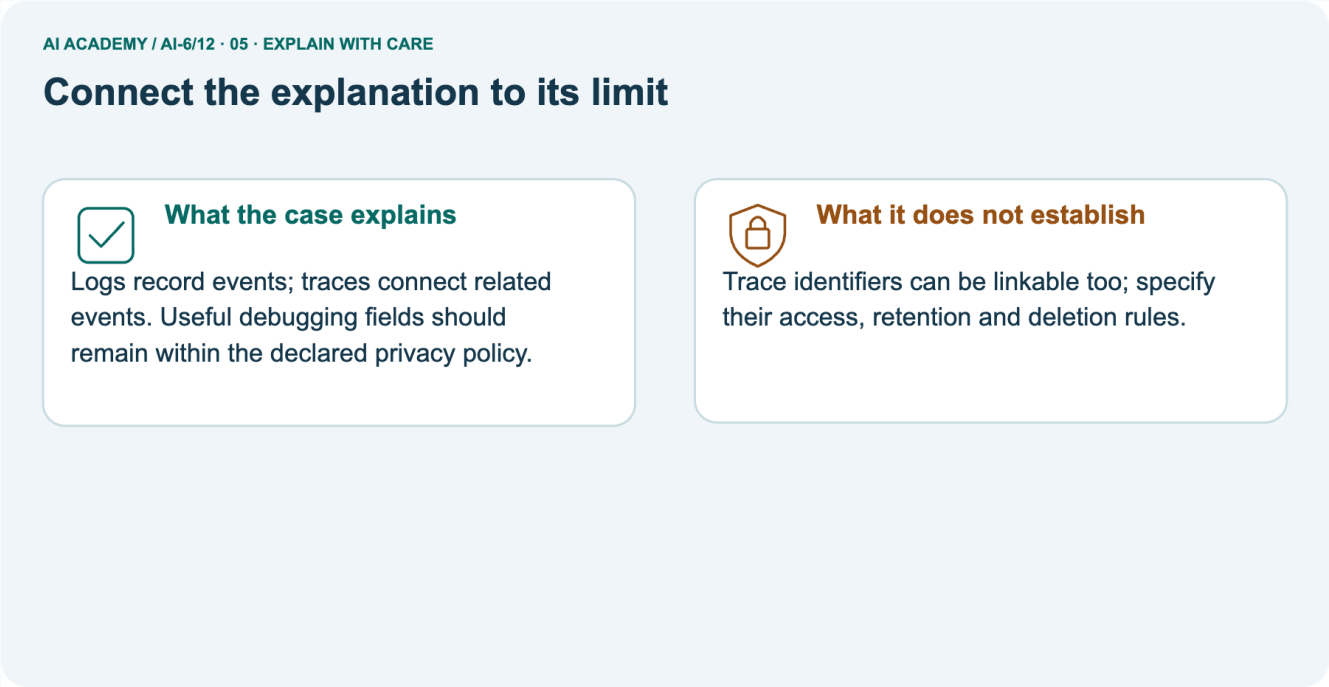
Original case: Keep allowed metadata; omit private question text and names. Changed condition: A developer logs the full learner question. Expected result: That additional field violates this logging policy.

Reason: Debugging convenience does not authorise storing more personal information.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



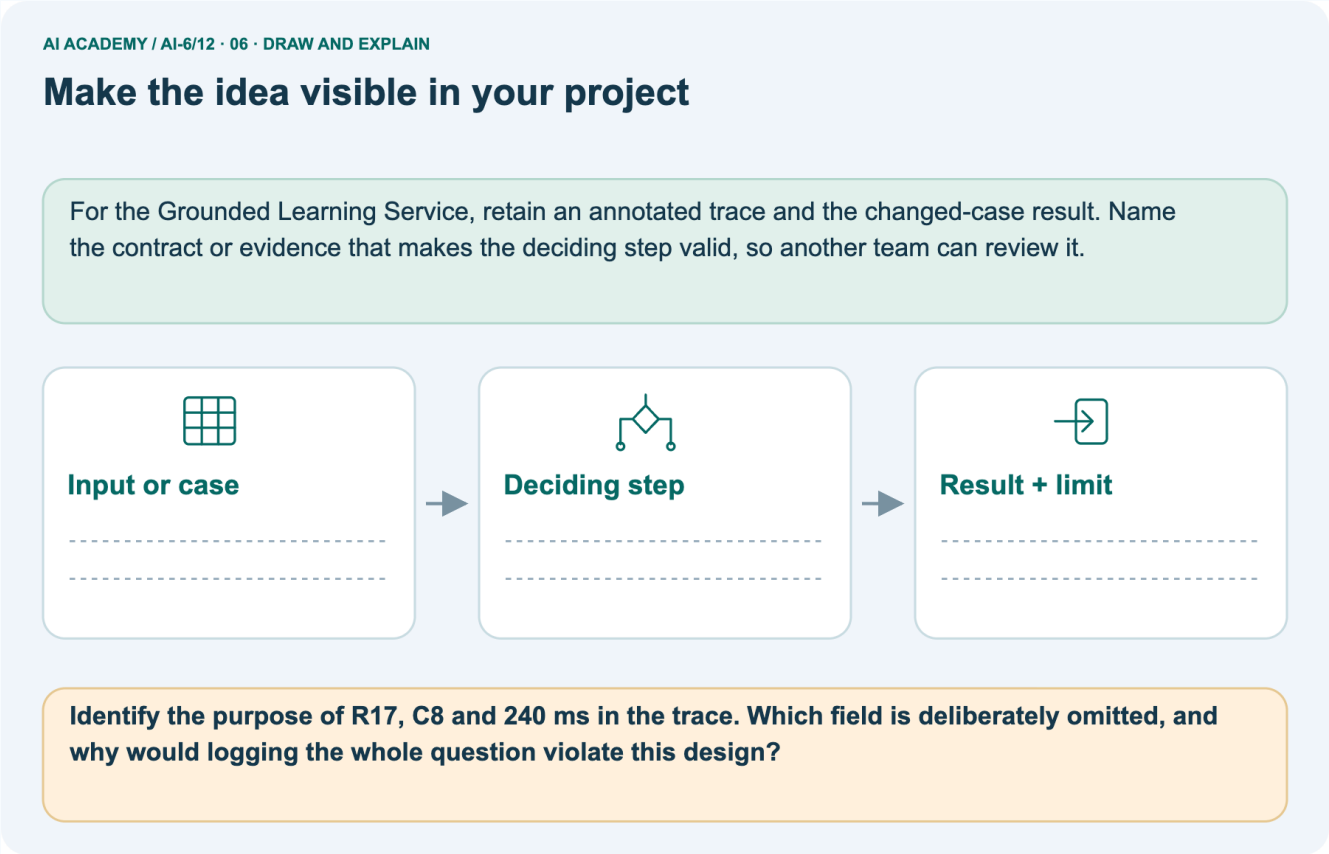
Follow the labelled picture with the matching explanation. Open the vector diagram

Logs record events; traces connect related events. Useful debugging fields should remain within the declared privacy policy. Limit: **Trace id**entifiers can be linkable too; specify their access, retention and deletion rules.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Identify the purpose of R17, C8 and 240 ms in the trace. Which field is deliberately omitted, and why would logging the whole question violate this design?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Logs traces and privacy: the case

02. Follow the deciding steps

03. Logs traces and privacy — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.